

The Bentil Group

A lab that CARES (i.e. Characterizing And REsearching Soft materials)!



[Home](#) [News](#) [Research](#) [Publications](#) [Funding](#) [People](#) [Facilities](#) [Collaborators](#) [Outreach](#)
[Teaching](#) [Donate](#) [Contact Us](#)

Advanced Blast Simulator (ABS) Commissioned at Iowa State University's CARES Lab

sbentil May 24, 2024 News

On May 23, 2024, commissioning shots were performed on the newly fabricated advanced blast simulator (ABS) at the CARES Lab, which is directed by Dr. Sarah A. Bentil at Iowa State University. "CARES" stands for "Characterizing And Rearching Soft materials".

The Bentil Group

The Bentil Group is on YouTube!

Visit The Bentil Group's
YouTube channel for more
videos: **[Click here to enter
the YouTube Channel for
The Bentil Group](#)**

Featured Equipment:
Advanced Blast
Simulator (ABS)



Advanced Blast Simulator (ABS) at Iowa State University

The ABS is oxyacetylene-driven, located in a newly renovated biosafety level 2 laboratory, and will support blast-induced injury (or damage) research. The length of the ABS is 43-ft (13.11 m). A 42-in. x 36-in. (1.07 m x 0.92 m) door is located on both sides of the ABS for sample insertion in the test section. The test section has a 4-ft x 4-ft (1.22 m x 1.22 m) cross-section.

*Collaborators interested in blast exposure experiments, using **biological and/or nonbiological samples** placed inside the ABS, should email Dr. Bentil (sbentil@iastate.edu).*



The ABS is oxyacetylene-driven, located in a newly renovated biosafety level 2 (BSL2) laboratory, and is designed to support blast-induced research using biological and non-biological samples. Click on the image of the ABS to learn more about the equipment.

Oxy-acetylene shock tube ignition



Oxy-acetylene Sh
The Bentil Group The Be...



00:00

00:28

Before and After: .177 cal pellet impacting a pig's brain



Advanced Blast Simulator (ABS) at Iowa State University

The Advanced Blast Simulator (ABS) was funded by the Office of Naval Research (ONR) through **PANTHER** (**P**hysics-**b**Ased **N**eutralization of **T**hreats to **H**uman tissu**E**s and o**R**gans).

← Interdisciplinary Biological Sciences Symposium: Spring 2024

Society for Experimental Mechanics (SEM): June 2024 →

00:0000:46



Ballistic Gel



.357 cal Big Bore
The Bentil Group The Be...



00:0001:22

Featured Member



Congratulations, Ling, on your paper with Jack being accepted for publication in the *Journal of the Mechanical Behavior of Biomedical Materials*! **Click here to read the paper.**

Featured Member



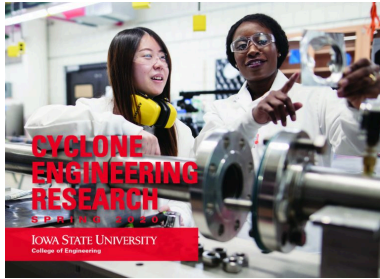
Congratulations, Jenny, on your review paper being accepted for publication in the Frontiers in Neurology Journal! [Click here to read the paper.](#)

Featured Member



Congratulations, Laura, on your review paper being accepted for publication in the Health Sciences Review Journal! [Click here to read the paper.](#)

Featured Member



Congratulations, Ling, on being featured on the March 2020 cover of the Cyclone Engineering Research Newsletter! [Click here to read more.](#)

Featured Member



Congratulations, Oluwaponmile, on your paper being accepted for publication in IEEE Transactions on Magnetics! [Click here to read the paper.](#)

Announcements

March 2026

March is **Brain Injury Awareness Month**. The theme for the 2026 campaign is **My Brain Injury Journey**. [Learn more...](#)

January 2026

Owen joins The Bentil Group as an undergraduate research assistant. Welcome!

June 2025

Dr. Bentil hosts the Grand Opening of The CARES Lab. CARES stands for Characterizing And REsearching Soft materials. Dr. Bentil hosts a two-day short course titled "Basics of Blast Physics and Simulation for Injury Studies". Dave Ritzel (Dyn-FX Consulting), the blast physicist who patented the "Advanced Blast Simulator", was the instructor for the course. [Click here to read more.](#)

March 2025

March is **Brain Injury Awareness Month**. The theme for the 2025 campaign is **My Brain Injury Journey**. [Learn more...](#)

February 2025

Jake joins The Bentil Group as an undergraduate research assistant. Welcome!

October 2024

Ian and Matthew join The Bentil Group as undergraduate research assistants. Welcome!

June 2024

Dulce presented her work at the Society for Experimental Mechanics' (SEM) Annual Conference and Exposition.

[Click here to read more.](#)

May 2024

Advanced Blast Simulator (ABS) is commissioned at Iowa State University. [Click here to learn more about the ABS.](#)

March 2024

March is **Brain Injury Awareness Month**. The theme for the 2024 campaign is **My Brain Injury Journey**.

[Learn more...](#)

Carley joins The Bentil Group as a Ph.D. student. Welcome!

February 2024

Ben joins The Bentil Group as an undergraduate research assistant. Welcome!

January 2024

Caden and Phuong join The Bentil Group as

undergraduate research
assistants. Welcome!

Past Announcements

[Read more...](#)

Featured Member



***Congratulations, Ling, on
being featured on the May
2019 cover of the Dimensions
Newsletter! [Click here to read
more.](#)***

Featured Member



Congratulations, Anastacia, on your paper with Ling, Sarah, and Jack being accepted for publication in the Journal of the Mechanical Behavior of Biomedical Materials! [Click here to read the paper.](#)

Featured Member



Congratulations, Ling, on your paper with Jack being accepted for publication in the Journal of the Mechanical Behavior of Biomedical

Materials! [Click here to read the paper.](#)

Anna Buchholz
provides a shock tube
tutorial

Shock Tube Tutor

The Bentil Group The Be...

00:00

03:28

Tim Beavers
introduces BrulSer
(Brain Impact
Simulator)

Tim J. Beavers int

The Bentil Group The Be...

00:00

03:14

Recent Posts

[Basics of Blast Physics and
Simulation for Injury Studies
\(2025\)
Society for Experimental
Mechanics \(SEM\): June 2024](#)

Advanced Blast Simulator
(ABS) Commissioned at Iowa
State University's CARES Lab
Interdisciplinary Biological
Sciences Symposium: Spring
2024

Categories

[Announcements](#)
[News](#)
[Open Positions](#)
[Outreach](#)
[Presentations](#)

Archives

[April 2025](#)
[June 2024](#)
[May 2024](#)
[April 2024](#)
[February 2024](#)
[April 2023](#)
[September 2022](#)
[June 2022](#)
[April 2022](#)
[April 2021](#)
[January 2021](#)
[September 2020](#)
[June 2020](#)
[April 2020](#)
[March 2020](#)
[October 2019](#)
[September 2019](#)
[June 2019](#)
[May 2019](#)
[April 2019](#)
[March 2019](#)

January 2019
December 2018
November 2018
July 2018
June 2018
April 2018
September 2017
August 2017
July 2017
May 2017
April 2017
January 2017
August 2016

Home

News

Research

Overview

Publications

Funding

People

Sarah A. Bentil, Ph.D.

Current Members

Former Members

The Bentil Group Photos

Facilities

Collaborators

Outreach

Teaching

Donate

Contact Us

THE BENTIL GROUP

Search